

FMFSTLT / TURBOTEST Public-Data Reproduction: Preliminary Results

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GitHub Repo: <https://github.com/g-nitin-1/FMFSTLT.git>

Objective. We are reproducing the TURBOTEST pipeline from public M-Lab data using a paper-faithful exact-public rebuild. The current milestone was to complete the Stage 1 throughput regressor, rebuild the Stage 2 stop/continue pipeline to match the paper semantics, and obtain preliminary quantitative results before the full epsilon sweep on GPU.

Completed engineering milestones.

- Rebuilt the exact-public dataset from normalized dense 100 ms tensors with 13 features per bucket.
- Completed Stage 1 XGBoost training and full scoring on train/val/test/robustness.
- Corrected Stage 2 label semantics to a *permanent-safe suffix oracle* with monotonic post- t^* stop labels.
- Rewrote Stage 2 data materialization to use *full-history inputs* and *500 ms decision stride*, instead of the earlier incorrect fixed 2 s window reuse.
- Validated the rewritten Stage 2 trainer with smoke/probe runs; WSL CUDA is available on the RTX 5080 laptop GPU and is ready for the full run.

Stage 1 quantitative results. Stage 1 trained on **20.7M** train windows and **2.30M** validation windows. The best boosting round was **1499**, so early stopping did not trigger.

Subset	Count	RMSE	MAE	Mean Rel. Error
Train	20.72M	47.76	19.62	0.373
Val	2.30M	52.48	19.88	0.370
Test	1.14M	46.20	18.55	0.353
Robustness	3.85M	173.52	35.23	0.341

These numbers show that the Stage 1 regressor is internally consistent across train/val/test. Robustness RMSE is much larger, which suggests heavier outliers, but the mean relative error remains close to the in-distribution test split.

Stage 2 dataset status at $\epsilon = 10\%$. We have already materialized the paper-faithful Stage 2 dataset using the corrected oracle, full-history sequences, and 500 ms decisions.

Subset	Tests	Decisions	Stop+ Rate	Oracle Stop/Test
Train	717k	13.45M	0.537	0.796
Val	79.7k	1.50M	0.537	0.796
Test	39.9k	750k	0.516	0.797
Robustness	132k	2.49M	0.517	0.795

This means that under a 10% tolerance, roughly **79.6%** of tests admit a permanent-safe early stop, and about **51–54%** of decision points are labeled as safe-to-stop after the oracle transition.

Interpretation. The main result so far is that the reproduction is no longer just infrastructure-complete: Stage 1 is fully trained and scored at scale, and Stage 2 has been corrected to match the paper’s intended semantics (monotonic oracle labels, full-history Transformer inputs, and 500 ms decision cadence). The preliminary numbers suggest that the Stage 1 regressor is good enough to support a meaningful Stage 2 early-exit policy study.

Immediate next step. The remaining work is to launch full **GPU-based Stage 2 training** in WSL CUDA for the epsilon sweep $\{5, 10, 15, 20, 25, 30, 35\}$, then report final stop/continue quality, stopping-time savings, and error under the learned early-exit policy.